



Evaluation Board for 12-/10-Bit, 8 Channel Parallel ADCs with a Sequencer

EVAL-AD7938/39/38-6CB

FEATURES

Full-Featured Evaluation Board for the AD7938/39/38-6

Eval-Control Board Compatible

Stand Alone Capability

On-Board Analog Buffering and Voltage Reference

On-Board Single Ended to Differential Conversion

Various Linking Options

PC Software for Control and Data Analysis when used with Evaluation Board Controller

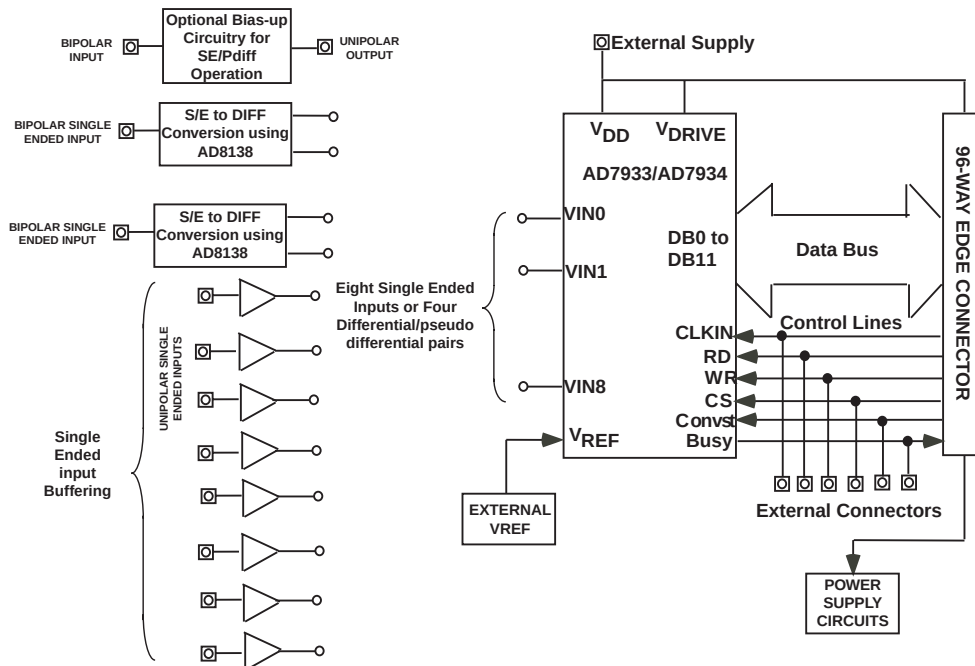
INTRODUCTION

This Technical Note describes the evaluation board for the AD7938/39/38-6 which are 12- & 10-bit, high speed, low power successive-approximation ADCs. These parts operate from a single 2.7V to 5.25V power supply and feature throughput rates up to 1.5MSPS. The AD7938/39/38-6 feature 8 analog input channels with a channel sequencer that allows a preprogrammed selection of channels to be converted sequentially. They can operate with either 8 Single-Ended, 4 Fully differential or 4 Pseudo differential analog inputs. Full details on the AD7938/39/38-6 are available in the AD7938/39/38-6 data sheet which is available from Analog Devices and should be consulted in conjunction with this Technical Note when using the Evaluation Board.

On-board components include an AD780 which is a pin programmable +2.5 V or +3 V ultra high precision bandgap reference, one AD713 quad op-amp, one AD8022 dual op-amp, six AD8021 single op-amps, two AD8138 differential amplifiers and a P174FCT digital buffer. Various link options are explained in detail in Table I.

This board has a 96-way connector which is compatible with the Eval-Control Board, which is also available from Analog Devices. External sockets are provided for a number of signals including the VREF input; the Analog inputs and the digital inputs and outputs.

FUNCTIONAL BLOCK DIAGRAM



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EVAL-AD7938CB

Link Options:

There are 60 link options which must be set for the required operating setup before using the evaluation board. The functions of the options are outlined below.

Table 1. Link Option Functions

Link No.	Function
LK1	This link option selects the source of the VDD +5 V supply used to supply the AD8138 differential amplifiers.
	In position "A", the VDD is supplied from the Eval-Control Board.
	In position "B", the VDD must be supplied from an external source via the power connector J3.
LK2	This link option selects the source of the VSS -5 V supply used to supply the AD8138 differential amplifiers.
	In position "A", the VSS is supplied from the Eval-Control Board.
	In position "B", the VSS must be supplied from an external source via the power connector J3.
LK3	This link option selects the source of the VDD supply for the AD7938/39/38-6 ADC.
	In position "A", VDD is supplied from the Eval-Control Board.
	In position "B", VDD must be supplied from an external source via J2.
LK4	This link option selects the source of the VDRIVE supply for the AD7934/AD7933 ADC digital interface.
	In position "A", VDRIVE is tied to VDD.
	In position "B", VDRIVE is supplied via the Eval Control board.
	In position "C", VDRIVE must be supplied from an external source via J4.
LK5	This link selects the source of the $\overline{WR}$ input to the ADC.
	In position "A", $\overline{WR}$ is supplied by the Eval Control Board.
	In position "B", $\overline{WR}$ must be supplied from an external source via P16.
LK6	This link selects where the BUSY output from the ADC appears.
	In position "A", the BUSY output may be read by the Eval-Control Board.
	In position "B", the BUSY may be read via the external socket, P15.
LK7	This link selects the source of the $\overline{CONVST}$ input to the ADC.
	In position "A", $\overline{CONVST}$ is supplied by the Eval Control Board.
	In position "B", $\overline{CONVST}$ must be supplied from an external source via P14.
LK8	This link selects the source of the $\overline{RD}$ input to the ADC.
	In position "A", $\overline{RD}$ is supplied by the Eval Control Board.
	In position "B", $\overline{RD}$ must be supplied from an external source via P13.
LK9	This link selects the source of the $\overline{CS}$ input to the ADC.
	In position "A", $\overline{CS}$ is supplied by the Eval Control Board.
	In position "B", $\overline{CS}$ must be supplied from an external source via P12.
LK10	This link selects the state of the $W/\overline{B}$ input.

	In position "A", $W/\overline{B}$ is tied low.
	In position "B", $W/\overline{B}$ is tied high.
LK11	This link selects the source of the CLKIN input to the ADC.
	In position "A", CLKIN is supplied by the Eval Control Board.
	In position "B", CLKIN must be supplied from an external source via P11.
LK12	If an external reference is being used, this link should be placed in position "A".
	If the internal reference is being used, this link should be placed in position "B".
LK13	This link selects the input to Vin4 pin on the ADC
	In position "A" the input is coming from VIN4 ,either in singled ended mode or as one channel of a differential pair.
	In position "B" the input to the ADC is tied to ground.
LK14	This link selects the input to Vin0 pin on the ADC
	In position "A" the input is coming from VIN0 ,either in singled ended mode or as one channel of a differential pair.
	In position "B" the input to the ADC is tied to ground.
LK15	This link selects the input to Vin1 pin on the ADC
	In position "A" the input is coming from VIN1 ,either in singled ended mode or as one channel of a differential pair.
	In position "B" the input to the ADC is tied to ground.
LK16	This link selects the input to Vin2 pin on the ADC
	In position "A" the input is coming from VIN2 ,either in singled ended mode or as one channel of a differential pair.
	In position "B" the input to the ADC is tied to ground.
LK17	This link selects the analog input to be applied to the buffer, U8, which is used when operating in single ended mode and when using VIN3/VIN7.
	If the single ended input applied to VIN3/VIN7 is Unipolar, this link should be in position "A", and the analog input should be applied at P10 with LK33 in position "B".
	If the single ended input applied to VIN3/VIN7 is Bipolar, this link should be in position "B" and the analog input should be biased up using the bias circuit by applying it to P18.
LK18	This link option adds a 50 Ω termination to AGND at the VIN3/VIN7 socket (P10) for the single ended input.
	This link should be inserted if a 50 Ω termination is required on the analog input.
LK19	This link is used to choose between a single ended analog input or a fully differential pair.
	If the analog input applied to VIN2/VIN6 is a single ended analog input then LK19 should be in position "B" as this input will be applied to the opamp buffer.
	If the analog input applied to VIN2/VIN6 is part of a differential pair with VIN3/VIN7, then LK19 should be in position "A" as this input will be applied to the AD8138 differential amplifier (U7).
LK20	This link selects the analog input to be applied to the buffer, U6, which is used when operating in single ended mode or pseudo differential mode and when using VIN2/VIN6.
	If the single ended input applied to VIN2/VIN6 is Unipolar, this link should be in position "A", and the analog input should be applied at P7 with LK19 in position "B".
	If the single ended input applied to VIN2 is Bipolar, this link should be in position "B" and the analog input should be biased up using the bias circuit by applying it to P18.
LK21	This link selects the analog input to be applied to the buffer, U5, which is used when operating in single ended mode and when using VIN1/VIN5.

	If the single ended input applied to VIN1/VIN5 is Unipolar, this link should be in position "A", and the analog input should be applied at P3 with LK34 in position "A".
	If the single ended input applied to VIN1 is Bipolar, this link should be in position "B" and the analog input should be biased up using the bias circuit by applying it to P18.
LK22	This link option adds a 50 Ω termination to AGND at the VIN1/VIN5 socket (P3) for the single ended input.
	This link should be inserted if a 50 Ω termination is required on the analog input.
LK23	This link option adds a 50 Ω termination to AGND at the VIN2/VIN6 socket (P7) for the single ended input.
	This link should be inserted if a 50 Ω termination is required on the analog input.
LK24	This link selects the input to the VIN- input of the AD8138 differential amplifier (U4).
	If the user applies a fully differential signal to VIN0 and VIN1 or VIN4 and VIN5, and only requires buffering of this signal before its applied to the ADC, then LK24 should be in position "A".
	If the user is applying a single ended input to VIN0/VIN4 and requires the use of the AD8138 to perform single ended to differential conversion, then this link should be in position "B". In this case, no input is applied to VIN1/VIN5.
LK25	This link selects the input to the VIN- input of the AD8138 differential amplifier (U7).
	If the user applies a fully differential signal to VIN2 and VIN3 or VIN6 and VIN7, and only requires buffering of this signal before its applied to the ADC, then Link 16 should be in position "A".
	If the user is applying a single ended input to VIN2/VIN6 and requires the use of the AD8138 to perform single ended to differential conversion, then this link should be in position "B". In this case, no input is applied to VIN3.
LK26	This link option adds a 50 Ω termination to AGND at the VIN1/VIN5 socket (P1) for the single ended input.
	This link should be inserted if a 50 Ω termination is required on the analog input.
LK27	This link is used to choose between a single ended analog input or a fully differential pair.
	If the analog input applied to VIN0/VIN4 is a single ended analog input then LK27 should be in position "A" as this input will be applied to the opamp buffer.
	If the analog input applied to VIN0/VIN4 is part of a differential pair with VIN1/VIN5, then LK27 should be in position "B" as this input will be applied to the AD8138 differential amplifier (U4).
LK28	This link selects the analog input to be applied to the buffer, U2, which is used when operating in single ended mode or pseudo differential mode and when using VIN0/VIN4.
	If the single ended input applied to VIN0/VIN4 is Unipolar, this link should be in position "A", and the analog input should be applied at P1 with LK27 in position "A".
	If the single ended input applied to VIN0 is Bipolar, this link should be in position "B" and the analog input should be biased up using the bias circuit by applying it to P18.
LK29	This link option selects the source of the VIN0/VIN4 analog input to be applied to the ADC.
	In position "A", a single ended, buffered signal is applied to VIN0/VIN4.
	In position "B", VIN0/VIN4 is supplied from the positive output of the AD8138 (U4) differential amplifier to provide half a differential pair with VIN1/VIN5.
	In position "C", an external VIN0/VIN4 is applied to the ADC via P4.
	In position "D", VIN0/VIN4 is tied to AGND. This should be done when VIN0/VIN4 is not being used and when power supplies are first applied to the board.
LK30	This link option selects the source of the VIN1/VIN5 analog input to be applied to the ADC.
	In position "A", a single ended, buffered signal is applied to VIN1/VIN5.
	In position "B", VIN1/VIN5 is supplied from the negative output of the AD8138 (U4) differential amplifier to provide half a differential pair with VIN0/VIN4.
	In position "C", an external VIN1/VIN5 is applied to the ADC via P5.
	In position "D", VIN1/VIN5 is tied to AGND. This should be done when VIN1/VIN5 is not being used and when power supplies are first applied to the board.

LK31	This link option selects the source of the VIN2/VIN6 analog input to be applied to the ADC.
	In position "A", a single ended, buffered signal is applied to VIN2/VIN6.
	In position "B", VIN2/VIN6 is supplied from the positive output of the AD8138 (U7) differential amplifier to provide half a differential pair with VIN3/VIN7.
	In position "C", an external VIN2/VIN6 is applied to the ADC via P6.
	In position "D", VIN2/VIN6 is tied to AGND. This should be done when VIN2/VIN6 is not being used and when power supplies are first applied to the board..
LK32	This link option selects the source of the VIN3/VIN7 analog input to be applied to the ADC.
	In position "A", a single ended, buffered signal is applied to VIN3/VIN7.
	In position "B", VIN3/VIN7 is supplied from the negative output of the AD8138 (U7) differential amplifier to provide half a differential pair with VIN2/VIN6.
	In position "C", an external VIN3/VIN7 is applied to the ADC via P9.
	In position "D", VIN3/VIN7 is tied to AGND. This should be done when VIN3/VIN7 is not being used and when power supplies are first applied to the board..
LK33	This link is used to choose between a single ended analog input or a fully differential pair.
	If the analog input applied to VIN3/VIN7 is part of a differential pair with VIN2/VIN6, then LK33 should be in position "A" as this input will be applied to the AD8138 differential amplifier.
	If the analog input applied to VIN3/VIN7 is a single ended analog input then LK33 should be in position "B" as this input will be applied to the opamp buffer.
LK34	This link is used to choose between a single ended analog input or a fully differential pair.
	If the analog input applied to VIN1/VIN5 is a single ended analog input then LK34 should be in position "A" as this input will be applied to the opamp buffer.
	If the analog input applied to VIN1/VIN5 is part of a differential pair with VIN0, then LK34 should be in position "B" as this input will be applied to the AD8138 differential amplifier.
LK35	This link selects the input to the VOCM pin (common mode input) of the AD8138 differential amplifier (U7).
	When in position "A", an external common mode input must be applied via P8.
	When in position "B", VREF is applied to the common mode input.
	When in position "C", VREF/2 is applied to the common mode input.
LK36	This link selects the input to the VOCM pin (common mode input) of the AD8138 differential amplifier (U4).
	When in position "A", an external common mode input must be applied via P2.
	When in position "B", VREF is applied to the common mode input.
	When in position "C", VREF/2 is applied to the common mode input.
LK37	This link selects the input to Vin3 pin on the ADC
	In position "A" the input is coming from VIN3, either in single ended mode or as one channel of a differential pair.
	In position "B" the input to the ADC is tied to ground.
LK38	This link selects whether the signal is applied to VIN0 or VIN4
	In position "A" the signal is applied to VIN0
	In position "B" the signal is applied to VIN4
LK39	This link selects the input to Vin5 pin on the ADC

	In position "A" the input is coming form VIN5 ,either in singled ended mode or as one channel of a differential pair.
	In position "B" the input to the ADC is tied to ground.
LK40	This link selects the input to Vin6 pin on the ADC
	In position "A" the input is coming form VIN6 ,either in singled ended mode or as one channel of a differential pair.
	In position "B" the input to the ADC is tied to ground.
LK41	This link selects the input to Vin7 pin on the ADC
	In position "A" the input is coming form VIN7 ,either in singled ended mode or as one channel of a differential pair.
	In position "B" the input to the ADC is tied to ground.
LK42	This link selects whether the signal is applied to VIN1 or VIN5
	In position "A" the signal is applied to VIN1
	In position "B" the signal is applied to VIN5
LK43	This link selects whether the signal is applied to VIN2 or VIN6
	In position "A" the signal is applied to VIN2
	In position "B" the signal is applied to VIN6
LK44	This link selects whether the signal is applied to VIN3 or VIN7
	In position "A" the signal is applied to VIN3
	In position "B" the signal is applied to VIN7
LK45	This link option adds a 50 Ω termination to AGND at the COM1 socket (P2) for the input to VOCM pin (common mode input) of the AD8138 differential amplifier (U4). .
	This link should be inserted if a 50 Ω termination is required on the input to VOCM pin (common mode input) of the AD8138 differential amplifier (U4) with LK36 in position "A".
LK46	This link option adds a 50 Ω termination to AGND at the COM2 socket (P2) for the input to VOCM pin (common mode input) of the AD8138 differential amplifier (U4). .
	This link should be inserted if a 50 Ω termination is required on the input to VOCM pin (common mode input) of the AD8138 differential amplifier (U4) with LK35 in position "A".
LK53	Always INSERT.
LK54	Always INSERT.
LK55	In position "A", REF is buffered to supply VREF to the Common mode voltage of the differential amplifier and to the bias up circuit.
	In position "B", REF is divided down by two then buffered to supply VREF/2 to the Common mode voltage of the AD8138 differential amplifier.
LK56	When VREF/2 is not being used, LK56 should be inserted so the inputs to U13C are not floating.
LK57	This link chooses the source of an external reference input.
	In position "A", an external reference should be applied via P17.
	In position "B", the output of the AD780 reference chip is applied to the VREF pin.
	In position "C", 3/4' s of the AD780 output is applied to the VREF pin. The resistors R35 and R52 can be changed if the user requires an alternative reference input.
LK58	This link determines whether the output of the AD780 reference chip is applied directly to LK57 or if it is divided down before being applied to LK57.

	In position "A", the output of the AD780 is applied to LK57.
	In position "B", the output of the AD780 is divided down before being applied to LK57.
LK59	This link option controls the program pin of the AD780 reference.
	When this pin is "Inserted", the AD780 output voltage is set to 3V.
	When this pin is "Removed", the AD780 output voltage is set to 2.5V.
LK60	This link option determines the source of the VCC (+5V) digital supply.
	When inserted, VCC is supplied via the Eval Control Board.
	When removed, VCC must be supplied to the external connector, J5.
LK61	This link selects the dc voltage to be applied to the bias up circuit which is used when a single ended, bipolar signal needs to be converted to a unipolar signal.
	In position "A", VREF is applied to the bias up circuit.
	In position "B", the dc input to the bias up circuit is tied to AGND. This should be done if the bias up circuit is not being used.
LK62	This link selects what portion of the VREF input is applied to the bias up circuit. The choice depends on the nature of the user's analog input signal.
	In position "A", 1/4 of the reference is applied to the bias up circuit
	In position "B", 1/2 of the reference is applied to the bias up circuit. If the bias up circuit is not used, then this link can be removed.
LK63	This link option adds a 50 Ω termination to AGND at the VIN S.E socket (P18) for the single ended input. This link should be inserted if a 50 Ω termination is required on the analog input.
LK64	This link option selects the source of the +12V power supply.
	In position "A", the +12 V is supplied by the Eval Control Board.
	In position "B", the +12 V must be supplied from an external source via J6.
LK65	This link option selects the source of the -12 V power supply.
	In position "A", the -12 V is supplied via the Eval Control Board.
	In position "B", the -12 V must be supplied from an external source via J6.
LK66	When the reference output of the AD780 is not being divided down, this link should be inserted so that the input of U10 is not floating.

Setup Conditions:

Care should be taken before applying power and signals to the Evaluation board to ensure that all link positions are as per the required operating mode. There are a few different modes in which to operate the evaluation board. The user can either operate the board with the eval control board or they can use it as a stand-alone board. The board can either be setup to accept 8 single ended inputs, 4 differential inputs, 4 pseudo differential inputs or 7 pseudo differential inputs. The link settings for the different modes of operation are detailed in Table II.

Table IIa Differential mode - Powered and Controlled by the Eval Control Board

The AD7938/39/38-6 can accept four fully differential analog input pairs. These can either be applied as two pairs to P1, P3 and P7, P10 or else single ended to differential conversion can be performed on a single ended input applied to P1 and P7. The following link positions are required for operating the Eval board in Differential mode. These are the link positions when the board is shipped. When the board is shipped, the assumption is that the user will be operating with the Eval-Control board so the links are set so that all power supplies and control signals are supplied by the Eval-control board. Initially, all analog inputs are tied to ground to ensure that on power up, they are not floating. The user must change these link positions depending on which analog inputs they are using.

Link No.	Position	Function
LK1	A	VDD for the AD8138 amplifiers is supplied by the eval control board.
LK2	A	VSS for the AD8138 amplifiers is supplied by the eval control board.
LK3	A	VDD for the AD7938/39/38-6 ADC is supplied by the eval control board.
LK4	B	VDRIVE is supplied by the eval control board.
LK5	A	$\overline{WB}$ is supplied by the eval control board.
LK6	A	BUSY is read by the eval control board.
LK7	A	$\overline{CONVST}$ is supplied by the eval control board.
LK8	A	$\overline{RD}$ is supplied by the eval control board.
LK9	A	$\overline{CS}$ is supplied by the eval control board.
LK10	B	ADC is set up to operate in WORD mode.
LK11	A	CLKIN is supplied by the eval control board.
LK12	A	An external reference is supplied to the VREFIN/OUT pin from the AD780.
LK13	A	If either a single ended input or half a differential input is applied to VIN4. LK13 should be in position B if VIN4 is not in use.
LK14	A	If either a single ended input or half a differential input is applied to VIN0. LK14 should be in position B if VIN0 is not in use.
LK15	A	If either a single ended input or half a differential input is applied to VIN1. LK15 should be in position B if VIN1 is not in use.
LK16	A	If either a single ended input or half a differential input is applied to VIN2. LK16 should be in position B if VIN2 is not in use.
LK17	B	The input to U8 is tied to Vbiased so it is not floating.
LK18	Inserted	On power up, the inputs to the opamps are tied to ground so they are not floating. Once a signal is applied to P10, LK18 can be removed if a 50ohm termination is not required.
LK19	A	If either a single ended input or half a differential input is applied to VIN2/VIN6 (P7).
LK20	B	The input to U6 is tied to Vbiased so it is not floating.
LK21	B	The input to U5 is tied to Vbiased so it is not floating.
LK22	Inserted	On power up, the inputs to the opamps are tied to ground so they are not floating. Once a signal is applied to P3, LK22 can be removed if a 50ohm termination is not required.
LK23	Inserted	On power up, the inputs to the opamps are tied to ground so they are not floating. Once a signal is applied to P7, LK23 can be removed if a 50ohm termination is not required.
LK24	B	If single ended to differential conversion is being performed on a single ended input applied to VIN0/VIN4 (P1). This link should be placed in position "A" if half a differential input is applied to P3
LK25	B	If single ended to differential conversion is being performed on a single ended input applied to VIN2/VIN6 (P7). This link should be placed in position "A" if half a differential input is applied to P10

LK26	Inserted	On power up, the inputs to the opamps are tied to ground so they are not floating. Once a signal is applied to P1, LK26 can be removed if a 50ohm termination is not required.
LK27	B	If either a single ended input or half a differential input is applied to VIN0/VIN4 (P1).
LK28	B	The input to U3 is tied to Vbiased so it is not floating.
LK29	D	On powerup, the analog input applied to VIN0 and/or VIN4 of the ADC is tied to AGND so that it is not floating. Once an analog input signal is applied to P1, VIN0/VIN4 of the ADC should be connected to the positive output of the AD8138 amplifier by placing LK29 in position B.
LK30	D	On powerup, the analog input applied to VIN1 and/or VIN5 of the ADC is tied to AGND so that it is not floating. Once an analog input signal is applied to P3, VIN1/VIN5 of the ADC should be connected to the positive output of the AD8138 amplifier by placing LK30 in position B.
LK31	D	On powerup, the analog input applied to VIN2 and/or VIN6 of the ADC is tied to AGND so that it is not floating. Once an analog input signal is applied to P7, VIN2/VIN6 of the ADC should be connected to the positive output of the AD8138 amplifier by placing LK31 in position B.
LK32	D	On powerup, the analog input applied to VIN3 and/or VIN7 of the ADC is tied to AGND so that it is not floating. Once an analog input signal is applied to P10, VIN3/VIN7 of the ADC should be connected to the positive output of the AD8138 amplifier by placing LK32 in position B.
LK33	Removed	If single ended to differential conversion is being performed on a single ended input applied to VIN2/VIN6 (P7). This link should be placed in position "A" if half a differential input is applied to P10
LK34	Removed	If single ended to differential conversion is being performed on a single ended input applied to VIN0/VIN4 (P1). This link should be placed in position "B" if half a differential input is applied to P3
LK35	B	VREF is applied to the VOCM pin of the AD8138 differential amplifier to set up the common mode voltage.
LK36	B	VREF is applied to the VOCM pin of the AD8138 differential amplifier to set up the common mode voltage.
LK37	A	If either a single ended input or half a differential input is applied to VIN3. LK37 should be in position B if VIN3 is not in use.
LK38	A&B	Both A&B are inserted in LK38 when applying half a differential input signal to VIN0 and VIN4
LK39	A	If either a single ended input or half a differential input is applied to VIN5. LK39 should be in position B if VIN5 is not in use.
LK40	A	If either a single ended input or half a differential input is applied to VIN6. LK40 should be in position B if VIN6 is not in use.
LK41	A	If either a single ended input or half a differential input is applied to VIN7. LK41 should be in position B if VIN7 is not in use.
LK42	A&B	Both A&B are inserted in LK42 when applying half a differential input signal to VIN1 and VIN5
LK43	A&B	Both A&B are inserted in LK43 when applying half a differential input signal to VIN2 and VIN6
LK44	A&B	Both A&B are inserted in LK44 when applying half a differential input signal to VIN3 and VIN7
LK45	Removed	This link should be inserted if a 50 Ω termination is required on the input to VOCM pin (common mode input) of the AD8138 differential amplifier (U4) with LK36 in position "A".
LK46	Removed	This link should be inserted if a 50 Ω termination is required on the input to VOCM pin (common mode input) of the AD8138 differential amplifier (U7) with LK35 in position "A".
LK53	Inserted	Always Insert
LK54	Inserted	Always Insert
LK55	A&B	The inputs to U13-A-&-B are not floating.
LK56	Inserted	The input to U13-C is not floating.
LK57	B	VREF is supplied by the AD780 reference chip.
LK58	A	The output of the AD780 supplies the reference to the ADC
LK59	Removed	The output of the AD780 is 2.5 V.
LK60	Inserted	VCC is supplied by the Eval Control Board
LK61	B	DC input to the bias up circuit is tied to AGND as it is not used in this mode
LK62	Removed	Bias up circuit not used in this mode

LK63	Inserted	Analog input to the bias up circuit is tied to AGND as it is not used in this mode.
LK64	A	+12 V is supplied by the Eval Control Board
LK65	A	-12 V is supplied by the Eval Control Board
LK66	Inserted	The output of the AD780 is not divided down so the input to U10 is tied to AGND.

Table IIb. Single Ended Mode - Powered and Controlled by the Eval Control Board

The AD7938/39/38-6 can operate with 8 single ended analog inputs. To operate the evaluation board in single ended mode, the link positions must be changed as shown in the table below. Initially, all analog inputs are tied to ground to ensure that on power up, they are not floating. The user must change these link positions depending on which analog inputs they are using.

Link No.	Position	Function
LK1	A	VDD for the AD8138 amplifiers is supplied by the eval control board.
LK2	A	VSS for the AD8138 amplifiers is supplied by the eval control board.
LK3	A	VDD for the AD7938/39/38-6 ADC is supplied by the eval control board.
LK4	B	VDRIVE is supplied by the eval control board.
LK5	A	$\overline{WR}$ is supplied by the eval control board.
LK6	A	BUSY is read by the eval control board.
LK7	A	$\overline{CONVST}$ is supplied by the eval control board.
LK8	A	$\overline{RD}$ is supplied by the eval control board.
LK9	A	$\overline{CS}$ is supplied by the eval control board.
LK10	B	ADC is set up to operate in WORD mode.
LK11	A	CLKIN is supplied by the eval control board.
LK12	A	An external reference is supplied to the VREFIN/OUT pin from the AD780.
LK13	A	If either a single ended input or half a differential input is applied to VIN4. LK13 should be in position B if VIN4 is not in use.
LK14	A	If either a single ended input or half a differential input is applied to VIN0. LK14 should be in position B if VIN0 is not in use.
LK15	A	If either a single ended input or half a differential input is applied to VIN1. LK15 should be in position B if VIN1 is not in use.
LK16	A	If either a single ended input or half a differential input is applied to VIN2. LK16 should be in position B if VIN2 is not in use.
LK17	A	A single ended, unipolar input must be supplied to VIN3/VIN7 (P10) and is buffered by U8.
LK18	Inserted	On power up, the inputs to the opamps are tied to ground so they are not floating. Once a signal is applied to P10, LK18 can be removed if a 50ohm termination is not required.
LK19	B	The analog input signal applied to P7 is applied to the single buffer (U6).
LK20	A	A single ended, unipolar input must be supplied to VIN2/VIN6 (P7) and is buffered by U6.
LK21	A	A single ended, unipolar input must be supplied to VIN1/VIN5 (P3) and is buffered by U5.
LK22	Inserted	On power up, the inputs to the opamps are tied to ground so they are not floating. Once a signal is applied to P3, LK22 can be removed if a 50ohm termination is not required.
LK23	Inserted	On power up, the inputs to the opamps are tied to ground so they are not floating. Once a signal is applied to P7, LK23 can be removed if a 50ohm termination is not required.
LK24	B	The negative input to the AD8138 is tied to AGND.
LK25	B	The negative input to the AD8138 is tied to AGND.
LK26	Inserted	On power up, the inputs to the opamps are tied to ground so they are not floating. Once a signal is applied to P1, LK26 can be removed if a 50ohm termination is not required.
LK27	A	The analog input signal applied to P1 is applied to the single buffer (U3).
LK28	A	A single ended, unipolar input must be supplied to VIN0/VIN4 (P1) and is buffered by U3.
LK29	D	On powerup, the analog input applied to VIN0 and/or VIN4 of the ADC is tied to AGND so

		that it is not floating. Once an analog input signal is applied to P1, VIN0/VIN4 of the ADC should be connected to the positive output of the AD8138 amplifier by placing LK29 in position A.
LK30	D	On powerup, the analog input applied to VIN1 and/or VIN5 of the ADC is tied to AGND so that it is not floating. Once an analog input signal is applied to P3, VIN1/VIN5 of the ADC should be connected to the positive output of the AD8138 amplifier by placing LK30 in position A.
LK31	D	On powerup, the analog input applied to VIN2 and/or VIN6 of the ADC is tied to AGND so that it is not floating. Once an analog input signal is applied to P7, VIN2/VIN6 of the ADC should be connected to the positive output of the AD8138 amplifier by placing LK31 in position A.
LK32	D	On powerup, the analog input applied to VIN3 and/or VIN7 of the ADC is tied to AGND so that it is not floating. Once an analog input signal is applied to P10, VIN3/VIN7 of the ADC should be connected to the positive output of the AD8138 amplifier by placing LK32 in position A.
LK33	B	The analog input signal applied to P10 is applied to a single buffer (U8).
LK34	A	The analog input signal applied to P3 is applied to the single buffer (U5).
LK35	A	The AD8138 is not being used so the common mode voltage needs to be tied to ground, with LK46 inserted
LK36	A	The AD8138 is not being used so the common mode voltage needs to be tied to ground, with LK45 inserted
LK37	A	If either a single ended input or half a differential input is applied to VIN3. LK37 should be in position B if VIN3 is not in use.
LK38	A&B	Both A&B are inserted in LK38 when applying a single ended input signal to VIN0 and VIN4
LK39	A	If either a single ended input or half a differential input is applied to VIN5. LK39 should be in position B if VIN5 is not in use.
LK40	A	If either a single ended input or half a differential input is applied to VIN6. LK40 should be in position B if VIN6 is not in use.
LK41	A	If either a single ended input or half a differential input is applied to VIN7. LK41 should be in position B if VIN7 is not in use.
LK42	A&B	Both A&B are inserted in LK42 when applying a single ended input signal to VIN1 and VIN5
LK43	A&B	Both A&B are inserted in LK43 when applying a single ended input signal to VIN2 and VIN6
LK44	A&B	Both A&B are inserted in LK44 when applying a single ended input signal to VIN3 and VIN7
LK45	Inserted	This link should be inserted if a 50 Ω termination is required on the input to VOCM pin (common mode input) of the AD8138 differential amplifier (U4) with LK36 in position "A".
LK46	Inserted	This link should be inserted if a 50 Ω termination is required on the input to VOCM pin (common mode input) of the AD8138 differential amplifier (U7) with LK35 in position "A".
LK53	Inserted	Always Insert
LK54	Inserted	Always Insert
LK55	A&B	The inputs to U13-A-&-B are not floating.
LK56	Inserted	The input to U13-C are not floating.
LK57	B	VREF is supplied by the AD780 reference chip.
LK58	A	The output of the AD780 supplies the reference to the ADC
LK59	Removed	The output of the AD780 is 2.5 V.
LK60	Inserted	VCC is supplied by the Eval Control Board
LK61	B	DC input to the bias up circuit is tied to AGND as it is not used in this mode
LK62	Removed	Bias up circuit not used in this mode
LK63	Inserted	Analog input to the bias up circuit is tied to AGND as it is not used in this mode.
LK64	A	+12 V is supplied by the Eval Control Board
LK65	A	-12 V is supplied by the Eval Control Board
LK66	Inserted	The output of the AD780 is not divided down so the input to U10 is tied to AGND.

Table IIc. Pseudo Differential mode - Powered and Controlled by the Eval Control Board

The AD7938/39/38-6 can accept four pseudo differential analog input pairs. The following links should be changed for Pseudo Differential mode, all other links are per Table IIa.

Link No.	Position	Function
LK1	Removed	No power supply is connected to VDD of the AD8138 differential amplifiers as they are not used in pseudo differential mode.
LK2	Removed	No power supply is connected to VSS of the AD8138 differential amplifiers as they are not used in pseudo differential mode.
LK17	A	If the DC portion of the pseudo differential input is to be buffered before being applied to VIN3/VIN7
LK19	B	The AC portion of the pseudo differential input applied to P7 is connected to the buffer U6 before being applied to VIN2/VIN6 of the ADC
LK20	A	The AC portion of the pseudo differential input applied to P7 is connected to the buffer U6 before being applied to VIN2/VIN6 of the ADC
LK21	A	If the DC portion of the pseudo differential input is to be buffered before being applied to VIN1/VIN5
LK27	A	The AC portion of the pseudo differential input applied to P1 is connected to the buffer U3 before being applied to VIN0/VIN4 of the ADC
LK28	A	The AC portion of the pseudo differential input applied to P1 is connected to the buffer U3 before being applied to VIN0/VIN4 of the ADC
LK29	D	On powerup, the analog input applied to VIN0 and/or VIN4 of the ADC is tied to AGND so that it is not floating. Once an analog input signal is applied to P1, VIN0/VIN4 of the ADC should be connected to the positive output of the AD8138 amplifier by placing LK29 in position A.
LK30	D	On powerup, the analog input applied to VIN1 and/or VIN5 of the ADC is tied to AGND so that it is not floating. Following power up, the user can either keep Vin- of the pseudo differential pair tied to AGND, or they can apply an external dc voltage either directly to P5 or they can apply it to P3 and buffer it before it is applied to VIN1/VIN5. Position B or C
LK31	D	On powerup, the analog input applied to VIN2 and/or VIN6 of the ADC is tied to AGND so that it is not floating. Once an analog input signal is applied to P7, VIN2/VIN6 of the ADC should be connected to the positive output of the AD8138 amplifier by placing LK31 in position A.
LK32	D	On powerup, the analog input applied to VIN3 and/or VIN7 of the ADC is tied to AGND so that it is not floating. Following power up, the user can either keep Vin- of the pseudo differential pair tied to AGND, or they can apply an external dc voltage either directly to P9 or they can apply it to P10 and buffer it before it is applied to VIN3/VIN7. Position B or C
LK33	B	If the DC portion of the pseudo differential input is to be buffered before being applied to VIN3/VIN7
LK34	A	If the DC portion of the pseudo differential input is to be buffered before being applied to VIN1/VIN5
LK35	A	The AD8138 is not being used so the common mode voltage needs to be tied to ground, with LK46 inserted
LK36	B	The AD8138 is not being used so the common mode voltage needs to be tied to ground, with LK45 inserted
LK45	Inserted	This link should be inserted if a 50 Ω termination is required on the input to VOCM pin (common mode input) of the AD8138 differential amplifier (U4) with LK36 in position "A".
LK46	Inserted	This link should be inserted if a 50 Ω termination is required on the input to VOCM pin (common mode input) of the AD8138 differential amplifier (U7) with LK35 in position "A".

Table II.d. Using the Bias Up Circuit in Single Ended or Pseudo Differential Mode

The bias up circuit can be used to bias up Single ended, bipolar signals to an appropriate dc voltage to make them unipolar to comply with the input requirements of the ADC. See table below for links changes, all other links remain as in table IIb.

Link No.	Position	Function
LK17	B	When operating in single ended mode, if the user is applying a bipolar input signal to the bias up circuit, it is applied to the buffer U8 via LK17.
LK20	B	When operating in single ended mode, if the user is applying a bipolar input signal to the bias up circuit, it is applied to the buffer U8 via LK20.
LK21	B	When operating in single ended mode, if the user is applying a bipolar input signal to the bias up circuit, it is applied to the buffer U8 via LK21.
LK28	B	When operating in single ended mode, if the user is applying a bipolar input signal to the bias up circuit, it is applied to the buffer U8 via LK28.
LK61	A	VREF is applied to the bias up circuit to setup the dc bias voltage.
LK62	A or B	Depending on what bias voltage is required.
LK63	Removed	Bias up circuit is used.

Table II.e. Stand-alone Operation - Single Ended, Unipolar Input Mode.

The AD7938/39/38-6CB evaluation board can be operated as a Stand-alone evaluation board. The user will have to supply all power supplies and control signals. The link options below are for Stand-alone operation with single ended unipolar inputs. Change the links appropriately for differential or pseudo differential inputs and if the bias up circuit is to be used. Any unused ADC analog input should be tied to AGND. To Write to or Read from the ADC databus in stand alone mode, use the 96-way connector. The pinout of this is shown in Table IV.

Link No.	Position	Function
LK1	A	VDD for the AD8138 amplifiers is supplied by the eval control board.
LK2	A	VSS for the AD8138 amplifiers is supplied by the eval control board.
LK3	B	An external supply is required to power the ADC via J2 (VDD_EXT)
LK4	A or B	VDRIVE is taken from VDD or an external VDRIVE should be applied via J3 (VDRIVE)
LK5	B	The $\overline{WR}$ control input should be applied via P16
LK6	B	The BUSY output should be read via P15
LK7	B	The $\overline{CONVST}$ control input should be applied via P14
LK8	B	The $\overline{RD}$ control input should be applied via P13
LK9	B	The $\overline{CS}$ control input should be applied via P12
LK10	B	ADC is set up to operate in WORD mode.
LK11	B	The CLKIN should be applied via P11
LK12	A	VREF is supplied by the AD780 reference chip.
LK13	A	If either a single ended input or half a differential input is applied to VIN4. LK13 should be in position B if VIN4 is not in use.
LK14	A	If either a single ended input or half a differential input is applied to VIN0. LK14 should be in position B if VIN0 is not in use.
LK15	A	If either a single ended input or half a differential input is applied to VIN1. LK15 should be in position B if VIN1 is not in use.
LK16	A	If either a single ended input or half a differential input is applied to VIN2. LK16 should be in position B if VIN2 is not in use.
LK17	A	A single ended unipolar input is applied to the buffer U8.
LK18	Inserted	On power up, the inputs to the opamps are tied to ground so they are not floating. Once a signal is applied to P10, LK18 can be removed if a 50ohm termination is not required.
LK19	B	The single ended, unipolar, analog input applied to P7 is connected to the buffer (U6).
LK20	A	A single ended, unipolar input is applied to buffer U6.
LK21	A	A single ended, unipolar input is applied to buffer U5.
LK22	Inserted	On power up, the inputs to the opamps are tied to ground so they are not floating. Once a signal is applied to P3, LK22 can be removed if a 50ohm termination is not required.

LK23	Inserted	On power up, the inputs to the opamps are tied to ground so they are not floating. Once a signal is applied to P7, LK23 can be removed if a 50ohm termination is not required.
LK24	B	The negative input to the AD8138 is tied to AGND as it is not used
LK25	B	The negative input to the AD8138 is tied to AGND as it is not used
LK26	Inserted	On power up, the inputs to the opamps are tied to ground so they are not floating. Once a signal is applied to P1, LK26 can be removed if a 50ohm termination is not required.
LK27	A	The single ended, unipolar, analog input applied to P1 is connected to the buffer (U3).
LK28	A	A single ended, unipolar input is applied to buffer U3.
LK29	D	On powerup, the analog input applied to VIN0 and/or VIN4 of the ADC is tied to AGND so that it is not floating. Once an analog input signal is applied to P1, VIN0/VIN4 of the ADC should be connected to the positive output of the AD8138 amplifier by placing LK29 in position A. Alternatively, the user can apply an analog input signal directly to P4, in this case LK29 should be in position C
LK30	D	On powerup, the analog input applied to VIN1 and/or VIN5 of the ADC is tied to AGND so that it is not floating. Once an analog input signal is applied to P3, VIN1/VIN5 of the ADC should be connected to the positive output of the AD8138 amplifier by placing LK30 in position A. Alternatively, the user can apply an analog input signal directly to P5, in this case LK30 should be in position C
LK31	D	On powerup, the analog input applied to VIN2 and/or VIN6 of the ADC is tied to AGND so that it is not floating. Once an analog input signal is applied to P7, VIN2/VIN6 of the ADC should be connected to the positive output of the AD8138 amplifier by placing LK31 in position A. Alternatively, the user can apply an analog input signal directly to P6, in this case LK31 should be in position C
LK32	D	On powerup, the analog input applied to VIN3 and/or VIN7 of the ADC is tied to AGND so that it is not floating. Once an analog input signal is applied to P10, VIN3/VIN7 of the ADC should be connected to the positive output of the AD8138 amplifier by placing LK32 in position A. Alternatively, the user can apply an analog input signal directly to P9, in this case LK32 should be in position C
LK33	B	The single ended, unipolar, analog input applied to P10 is connected to the buffer (U8).
LK34	A	The analog input signal applied to P3 is applied to the single buffer (U5).
LK35	A	The AD8138 is not being used so the common mode voltage needs to be tied to ground, with LK46 inserted
LK36	B	The AD8138 is not being used so the common mode voltage needs to be tied to ground, with LK45 inserted
LK37	A	If either a single ended input or half a differential input is applied to VIN3. LK37 should be in position B if VIN3 is not in use.
LK38	A&B	Both A&B are inserted in LK38 when applying a single ended input signal to VIN0 and VIN4
LK39	A	If either a single ended input or half a differential input is applied to VIN5. LK39 should be in position B if VIN5 is not in use.
LK40	A	If either a single ended input or half a differential input is applied to VIN6. LK40 should be in position B if VIN6 is not in use.
LK41	A	If either a single ended input or half a differential input is applied to VIN7. LK 40 should be in position B if VIN7 is not in use.
LK42	A&B	Both A&B are inserted in LK42 when applying a single ended input signal to VIN1 and VIN5
LK43	A&B	Both A&B are inserted in LK43 when applying a single ended input signal to VIN2 and VIN6
LK44	A&B	Both A&B are inserted in LK44 when applying a single ended input signal to VIN3 and VIN7
LK45	Inserted	This link should be inserted if a 50 Ω termination is required on the input to VOCM pin (common mode input) of the AD8138 differential amplifier (U4) with LK36 in position "A".
LK46	Inserted	This link should be inserted if a 50 Ω termination is required on the input to VOCM pin (common mode input) of the AD8138 differential amplifier (U7) with LK35 in position "A".
LK53	Inserted	Always Insert
LK54	Inserted	Always Insert
LK55	A&B	Bias up circuit is not used.

LK56	Inserted	The input to U13-C are not floating.
LK57	A or B	The user should apply an external reference via P17 or the reference is supplied by the AD780.
LK58		See definition Table 1
LK59		See definition Table 1
LK60	Remove	An external 5V supply for VCC should be applied via J5
LK61		See definition Table 1
LK62		See definition Table 1
LK63		See definition Table 1
LK64	B	An external +12 V supply should be applied via J6 to supply the opamps and the voltage reference
LK65	B	An external -12 V supply should be applied via J6 to supply the opamps and the voltage reference

INTERFACING THE EVALUATION BOARD TO THE EVAL CONTROL BOARD

Interfacing the Eval control board to the evaluation board is via a 96-way connector, J1. The pinout for the J1 connector is shown in Figure 1. Table III gives a description of the pins on the 96-way connector used to interface between the Eval-Control Board and the EVAL-AD7933/34CB, and Table IV gives its pin designations.

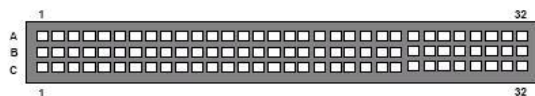


Figure 1. Pin Configuration for the 96-Way Connector J1

Pin	Description
D0-D11	Data Bit 0 to Data Bit 11. Three state TTL I/O pins used to read conversion data and to write data to the internal registers of the ADC. D11 is the MSB and D0 the LSB for the AD7938 and D2 the LSB for the AD7939.
SCLK0	Serial Clock. This continuous clock is connected to the CLKIN pin of the AD7938/39/38-6 via LK11 .
+5VD	Digital +5V supply. This is used to provide a digital supply to the board for the digital logic.
$\overline{\text{RD}}$	Read. This is an active low logic input connected to the $\overline{\text{RD}}$ pin of the AD7938/39/38-6 via LK8.
$\overline{\text{WR}}$	Write. This is an active low logic input connected to the $\overline{\text{WR}}$ pin of the AD7938/39/38-6 via LK5.
$\overline{\text{CS}}$	Chip Select. This is an active low logic input connected to the $\overline{\text{CS}}$ pin of the AD7938/39/38-6 via LK9.
FL0	Flag zero. This logic input is connected to the CONVST input of the AD7938/39/38-6 via LK7.
$\overline{\text{IRQ2}}$	Interrupt request 2. This is a logic output and is connected to the BUSY output of the AD7938/39/38-6.
DGND	Digital Ground. These lines are connected to the digital ground plane on the Evaluation board. This allows the user to provide a digital power supply via the connector along with other digital signals.
AGND	Analog Ground. These lines are connected to the Analog ground plane on the evaluation board.
AVDD	Analog +5V Supply. These lines are connected to the AVDD supply line on the evaluation board via LK1 to provide +5V to the AD8138 differential amplifiers. They are also connected to the VDD supply of the AD7938/39/38-6 via LK3

Pin	Description
AVSS	Analog -5V Supply. These lines are connected to the AVSS supply line on the evaluation board via LK2 to supply -5V to the AD8138 differential amplifiers.
$\pm 12\text{ V}$	$\pm 12\text{V}$ Supply. These lines are connected to the $\pm 12\text{V}$ supply lines on the evaluation board via LK64 & LK65 to supply the AD713, the AD8021s, AD8022 and the AD780.

Table IV. 96-Way Connector Pin Functions.

	ROW A	ROW B	ROW C
1			
2		D0	
3		D1	
4	DGND	DGND	DGND
5		D2	
6		D3	
7	SCLK0	D4	SCLK0
8	+5V	+5V	+5V
9	$\overline{\text{RD}}$	D5	$\overline{\text{WR}}$
10		D6	$\overline{\text{CS}}$
11		D7	
12	DGND	DGND	DGND
13		D8	
14		D9	
15		D10	
16	DGND	DGND	DGND
17	FL0	D11	$\overline{\text{IRQ2}}$
18			
19			
20	DGND	DGND	DGND
21	AGND	AGND	AGND
22	AGND	AGND	AGND
23	AGND	AGND	AGND
24	AGND	AGND	AGND
25	AGND	AGND	AGND
26	AGND	AGND	AGND
27		AGND	
28		AGND	
29	AGND	AGND	AGND
30	-12V	AGND	+12V
31	AVSS	AVSS	AVSS
32	AVDD	AVDD	AVDD

Note: The unused pins of the 96-way connector are not shown.

SOCKETS

There are **eighteen** SMB input sockets relevant to the operation of the AD7938/39/38-6 on this evaluation board. All of these sockets are used for applying an externally generated signal to the evaluation board. When operating the board with the Eval-Control Board the only necessary external sockets are used to supply the analog inputs to the ADC (i.e P1, P3, P7, P10). All of the other sockets are optional and if they are not used their signals are supplied by the Eval-Control Board. Most of these sockets are used when operating the board as a stand-alone unit, as all the signals required are supplied from external sources. The functions of these sockets are outlined in Table V.

Table V. Socket Functions

Socket	Function
P1	Vin0/Vin4. Sub-Miniature BNC socket for either a single ended unipolar analog input to be applied to the VIN0/VIN4 input of the ADC in single ended mode; or for a bipolar single ended analog input to be applied to the AD8138 for single ended to differential conversion in differential mode; or for half a differential signal to be applied to the AD8138 for differential signal buffering.
P2	COM1. Sub-Miniature BNC socket for the dc analog input to Vocm pin on the AD8138 differential amplifier
P3	Vin1/Vin5. Sub-Miniature BNC socket for either a single ended unipolar analog input to be applied to the VIN1/VIN5 input of the ADC in single ended mode; or for half a differential signal to be applied to the AD8138 for differential signal buffering.
P4	Ext_Vin0/Vin4. Sub-Miniature BNC socket for a signal to be applied directly to VIN0/VIN4 input of the ADC.
P5	Ext_Vin1/Vin5. Sub-Miniature BNC socket for a signal to be applied directly to VIN1/VIN5 input of the ADC.
P6	Ext_Vin2/Vin6. Sub-Miniature BNC socket for a signal to be applied directly to VIN2/VIN6 input of the ADC.
P7	Vin2/Vin5. Sub-Miniature BNC socket for either a single ended unipolar analog input to be applied to the VIN2/VIN5 input of the ADC in single ended mode; or for a bipolar single ended analog input to be applied to the AD8138 for single ended to differential conversion in differential mode; or for half a differential signal to be applied to the AD8138 for differential signal buffering.
P8	COM2. Sub-Miniature BNC socket for the dc analog input to Vocm pin on the AD8138 differential amplifier.
P9	Ext_Vin3/Vin7. Sub-Miniature BNC socket for a signal to be applied directly to VIN3/VIN7 input of the ADC

Socket	Function
P10	Vin3/Vin7. Sub-Miniature BNC socket for either a single ended unipolar analog input to be applied to the VIN3 input of the ADC in single ended mode; or for half a differential signal to be applied to the AD8138 for differential signal buffering.
P11	CLKIN. Sub-Miniature BNC socket for an external Clock input.
P12	CS. Sub-Miniature BNC socket for an external $\overline{CS}$ input.
P13	RD. Sub-Miniature BNC socket for an external $\overline{RD}$ input.
P14	CONVST. Sub-Miniature BNC socket for an external $\overline{CONVST}$ input.
P15	Busy. Sub-Miniature BNC socket for reading the Busy output.
P16	WR. Sub-Miniature BNC socket for an external $\overline{WR}$ input.
P17	VREF. Sub-Miniature BNC socket for an external VREF input.
P18	VIN S.E. Sub-Miniature BNC socket for the bipolar single ended or pseudo differential analog input to the bias-up circuit.

CONNECTORS

There are **six** connectors on the AD7938/39/38-6CB evaluation board as outlined in Table VI.

Table VI. Connector Functions

Connector	Function
J1	96-Way Connector for the Digital Interface & Power Supply connections.
J2	External VDD power connector for the ADC.
J3	External AVDD, AVSS & AGND power connector.
J4	External VDRIVE power connector.
J5	External Digital +5V power connector.
J6	External +12 V, -12 V & AGND power connector.

TEST POINTS

There are 13 test points on the inputs to the AD7938/39/38-6CB on the evaluation board. These enable the user to have easy access to these signals for probing, evaluation and debugging.

OPERATING WITH THE EVAL-CONTROL BOARD

The evaluation board can be operated in a stand alone mode or operated in conjunction with the EVAL-CONTROL BOARD. This Evaluation Controller Board is available from Analog Devices under the order entry "EVAL-CONTROL BRD2".

When interfacing the EVAL-AD7938/39/38-6CB directly to the Eval-Control Board, all supplies and control signals to operate the AD7938/39/38-6 board are provided by the EVAL-CONTROL BOARD.

Due to the nature of the DSP interface on the Eval-Control Board, AD7938/39/38-6 sampling rates greater than 800kSPS are not supported when interfacing this evaluation board directly to the Eval-Control Board.

Software to communicate with the Eval-Control Board and AD7938/39/38-6 is provided with the AD7938/39/38-6 evaluation board package. This Eval-Control Board will also operate with all Analog Devices evaluation boards which end with the letters CB in their title.

AD7938/39/38-6 SOFTWARE

Included in the EVAL-AD7938/39/38-6CB evaluation kit is a CD ROM which contains software for controlling and evaluating the performance of the AD7938/39/38-6 when it is operated with the Eval-Control Board. When the CD is inserted into the PC, an installation program will automatically begin. This program will install the evaluation software onto the user's machine and will also install the technical note for the evaluation board as well as the datasheet for the AD7938/39/38-6 and the technical note for the Eval-Control Board. All literature on the CD is in Adobe's Portable Documentation format (PDF) and will require Acrobat Reader™ to be viewed or printed. The user interface on the PC is a dedicated program written especially for the AD7938/39/38-6 when operating with the eval-control board.

SOFTWARE DESCRIPTION

The software that controls the Eval-Control Board and hence the AD7938/39/38-6 evaluation board has two main screens. The screen shown in Figure 2 is the screen which appears when the software is run. The main function of this screen is to allow the user to read a pre-determined number of samples from the evaluation board and display them in both the time and frequency domain. The screen can be divided into three sections.

The **Upper third** of the screen contains the control buttons, the menu bar, the busy status and selection windows. The **Control buttons** allow the user to take samples, reset board, to **exit** the program and to open the **Load Configuration window** to load a configuration file by clicking on **Device Select**. The **Control Reg** button is used to access the on-chip registers. The user can use the drop down menus to change between the different modes (i.e single ended, differential etc) but the user must use the control reg button in order to access the on chip sequence and shadow registers. This is explained in further detail below.

The **menu bar** consists of **File**, **Config**, **Channel** and **About**.

The 96-way connector on the EVAL-AD7938/39/38-6CB plugs directly into the 96-way connector on the Eval-Control Board. The Eval-Control Board provides all the supplies for the evaluation board. It is powered from a 12V AC transformer. Suitable transformers are available from Analog Devices as an accessory under the following part numbers:

EVAL-110VAC-US: For use in the U.S. or Japan

EVAL-220VAC-UK: For use in the U.K.

EVAL-220VAC-EU: For use in Europe

These transformers are also available from other suppliers including Digikey (U.S.) and Campbell Collins (U.K.).

Connection between the Eval-Control Board and the serial port of a PC is via a standard Centronics printer port cable which is provided as part of the Eval-Control Board package. Please refer to the manual which accompanies the Eval-Control Board for more details on this package.

The **File** menu allows the following:

Load Raw Data: Selecting this option allows the user to load data which had been saved by the software during a previous session.

Save Raw Data: Selecting this option allows the user to save the current set of sample data points. The data can be reloaded to the Eval-Control Board software at a later date or can be used by other programs for further analysis.

Save Binary Data: Selecting this option allows the user to save the current set of sample data points. The data is saved in binary format as a text file. This method can be useful for examining code flicker, looking for stuck bits etc.

Exit: Quits the program.

The **Config** drop down menu allows the user to setup certain operating conditions in the ADC control register i.e. Power management, output coding, internal or external reference and the analog input range.

The **Channel** drop down menu allows the user to choose their analog input type (single ended, differential or pseudo differential), which analog input to convert.

In order to use the on chip sequence please see the section below.

The **About** drop down menu gives the user information about the version of the software.

The **Busy status** indicates to the user when the evaluation board is busy.

The **Selection Windows** allow the user to change the sampling frequency and the number of samples to upload.

The **Codes/Volts** button determines whether the data is displayed in volts or codes.

The **Frequency** window displays the speed at which the part is running. This can be changed by the user. Please note that when using the sequencer the parts will run at the speed chosen by the user but, this speed will be divided down by the number of channels selected at any one time.

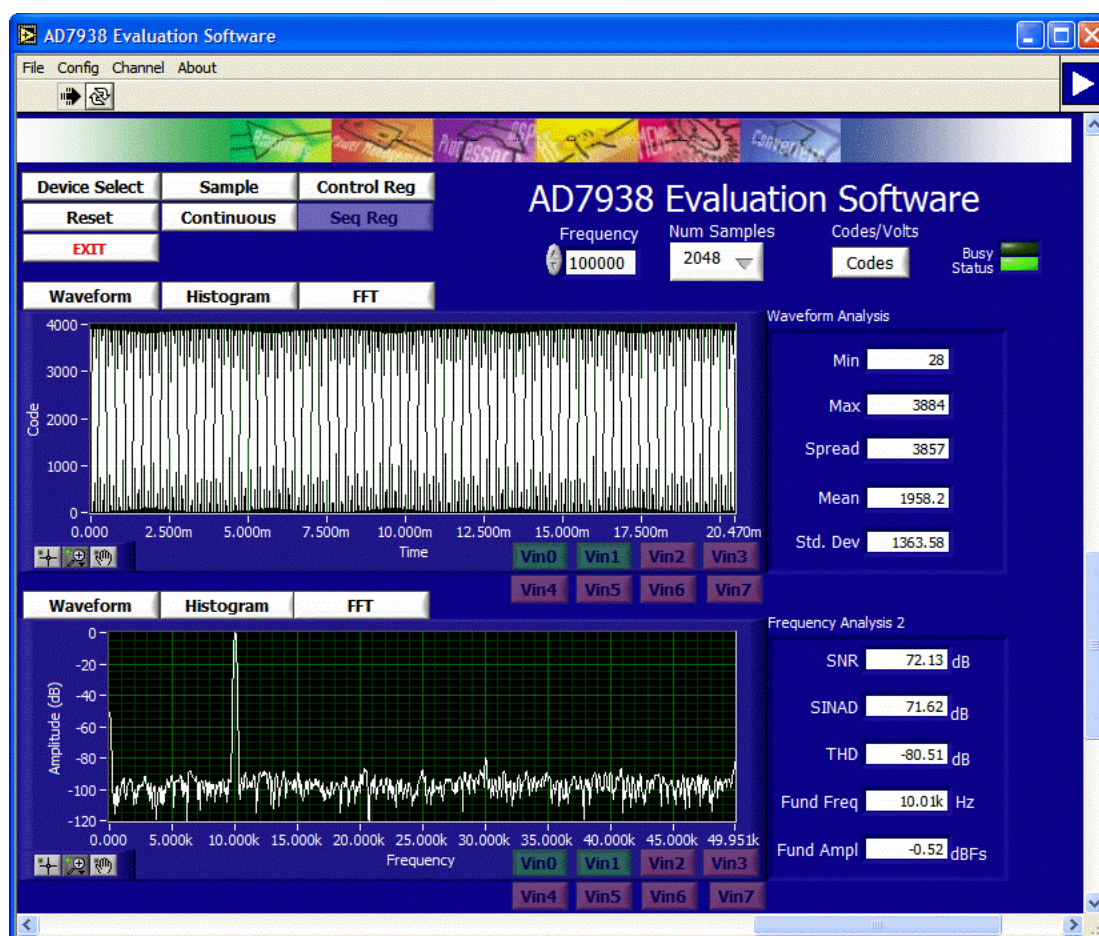


Figure 2. AD7938/39/38-6 Main Screen

The **middle third** of the screen is a Digital Storage Oscilloscope (DSO) which allows the user to display either a **Waveform, a Histogram or an FFT**. When samples are uploaded from the Eval-Control Board they are displayed here. The Samples can be displayed either as integer values or as voltages. At the bottom left of the graph are the zoom options. These allow the user to zoom in and out to get a closer look at a sample, if required. The right hand side of the middle section contains information about the samples taken like min/max position, the spread, the standard deviation and the mean. The **lower third** of the screen is also a Digital Storage Oscilloscope (DSO) which allows the user to display either a **Waveform, a Histogram or an FFT**. The FFT (the default option) is typically used when the user is concerned with examining the ADCs performance in the frequency domain while the Histogram will give an indication of the ADCs performance in response to dc inputs. The option displayed can be changed by clicking on the Waveform, Histogram and FFT buttons. The right hand side of the

lower section contains information about the samples taken e.g. ac specifications. Figure 2 shows a screen shot of the main screen.

Load Configuration Screen

The Load Configuration Screen is responsible for allowing the user to load the required configuration file for the evaluation board. A screenshot is shown in Figure 3. The configuration gives the software detailed information about the AD7938/39/38-6 Evaluation Board and the part connected to the Eval-Control Board such as the number of bits, the maximum sampling rate, output coding, maximum sampling rate, power supply requirements etc.

The configuration file also tells the software the names of the DSP program file which it should download to the Eval-Control Board. The Load Configuration Screen also allows the user to choose the sampling frequency and the number of samples to take.

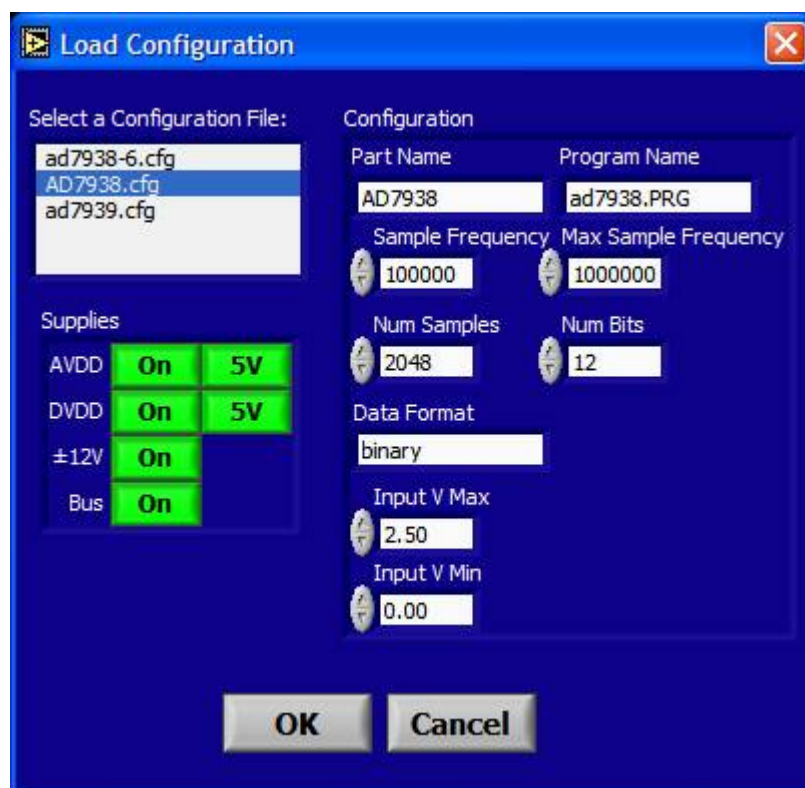


Figure 3. AD7938/39/38-6 Load Configuration Screen

In order to use the sequencer register function the user must firstly write to the control register. To do this the user should click on the **Control Reg** button on the screen. The control register panel, as in Figure 4, will open. See On Chip Registers section of the datasheet for details on using the control, sequence and shadow registers.



Figure 4: The Control Register

SETTING UP THE EVAL-CONTROL BOARD:

This section describes how the evaluation board, the Eval-Control Board and the software should be set up for the user to begin using the complete system.

The Eval-Control Board and Evaluation board should be connected together (via the 96-way connector). The power should be applied to the Eval-Control Board via a 12V AC Transformer. At this stage the red LED on the Eval-Control Board should be flashing which indicates that that Eval-Control Board is functional and ready to receive instructions. The software, which should have been installed should be loaded before the printer port cable is connected between the Eval-Control Board and the PC. This will

ensure that the printer port has been initialized properly. The printer port can then be connected between the PC and the Eval-Control Board.

Running the software

With the hardware set up the user is now in a position to use the software to control the Eval-Control Board and the AD7938/39/38-6 evaluation board.

When the software is run, the user should click on the Device Select control button. This will display the Load Configuration Window (as shown in Figure 3). The window on the top left of the set up window lists the available configuration files. The configuration files are text based files which contain information

about the particular evaluation board to be tested. The information covers the part name, number of samples to be taken, default and maximum sampling frequency, power supply settings etc. The configuration file also contains the name of the DSP program file which is to be downloaded to the Eval-Control Board. The user should select the relevant configuration file and click **OK**. The Eval-Control Board will be reset and the DSP program will be downloaded. When the download has been completed the power supply settings indicated in the configuration file are set and the user may hear some of the relays clicking. The selection windows (e.g. 'number of samples' and 'sampling frequency') will have been set to the default values specified by the configuration file. The user

is free to change these at will.

Taking Samples

When the user clicks Sample, the software will instruct the Eval-Control Board to take the required number of samples at the required frequency from the evaluation board.

The samples taken are then uploaded and displayed. An FFT and Histogram are also calculated and displayed. If the user clicks **Continuous** the software will repeat the process indefinitely until the user clicks **STOP**.

SOFTWARE CONFIGURATION FILES:

Software Configuration Files give the Eval-Control Board software information on how the software and hardware should perform. They contain information such as the name of the DSP program to download, the default and maximum sample frequencies, the number of samples to take and the power supply settings to use. A typical Software Configuration File (*.cfg) is shown in Listing 1.

EVAL-AD7938CB Schematic, Layout and Bill of Materials.

These items can be found on the following pages.

[EVAL-CONTROL BOARD]

partname:AD7938
programname:ad7938.PRG

samplefrequency:100000
maxsamplefrequency:1000000
samples:2048

+/-15V:on
dvdd:5:on
avdd:5:on
bus:on
;options 2scomp, binary
dataformat:binary
numberofbits:12
inputVmax:2.5
inputVmin:0
[endofconfig]

Listing 1. Typical Configuration File

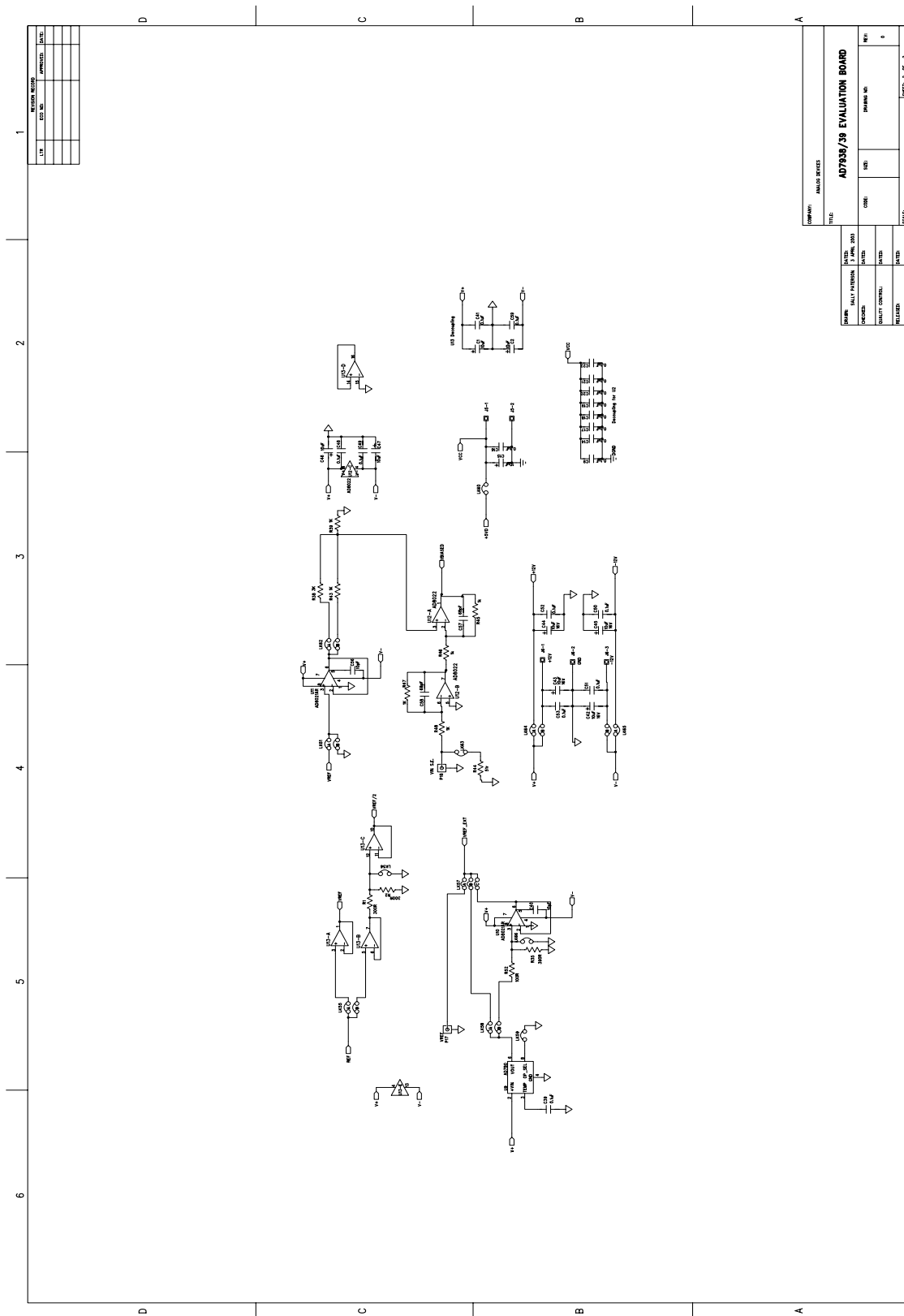


Figure 5. AD7938/39/38-6 Evaluation Board Circuit Diagram, Page 1

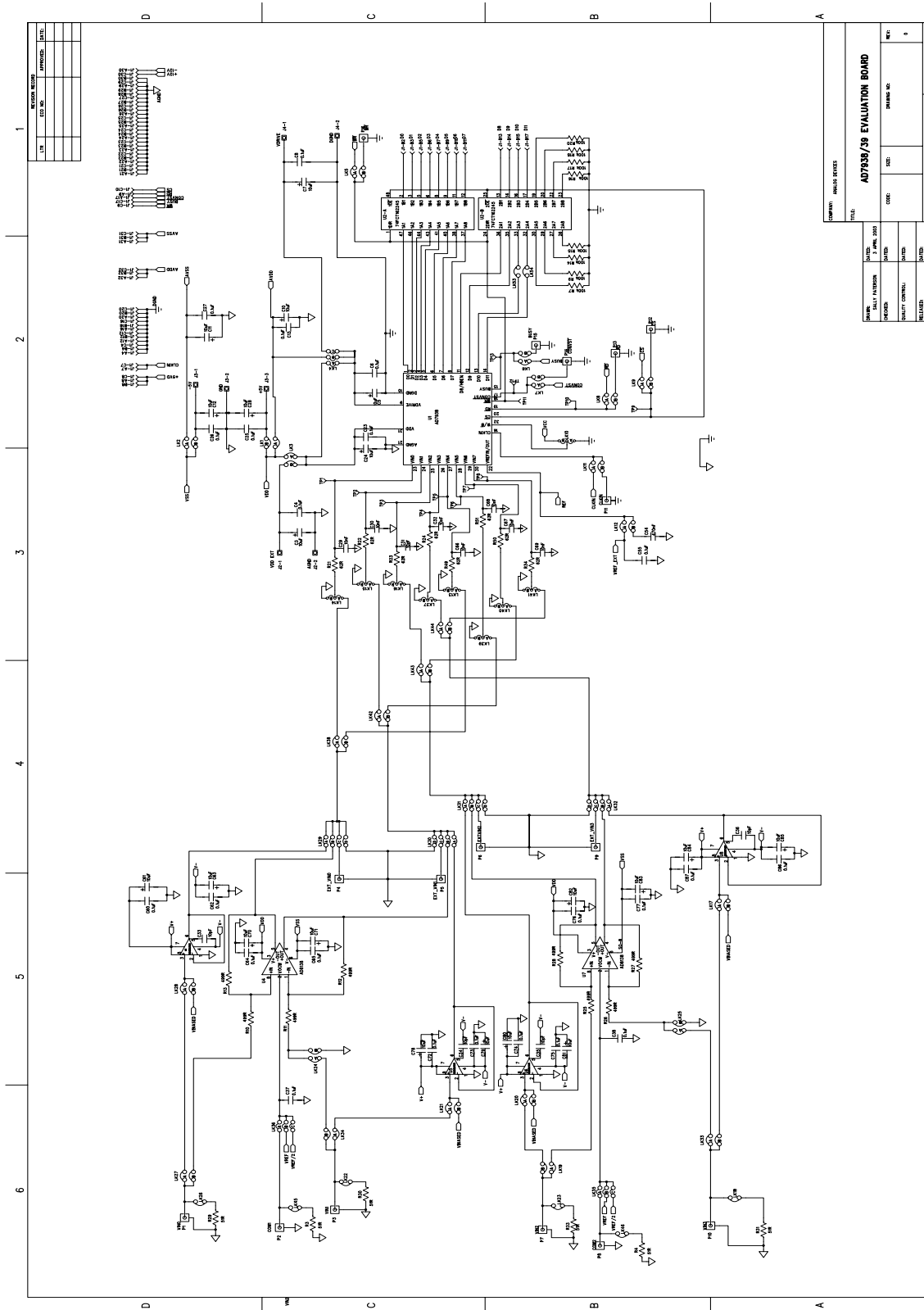


Figure 6. AD7938/39/38-6 Evaluation Board Circuit Diagram, Page 2

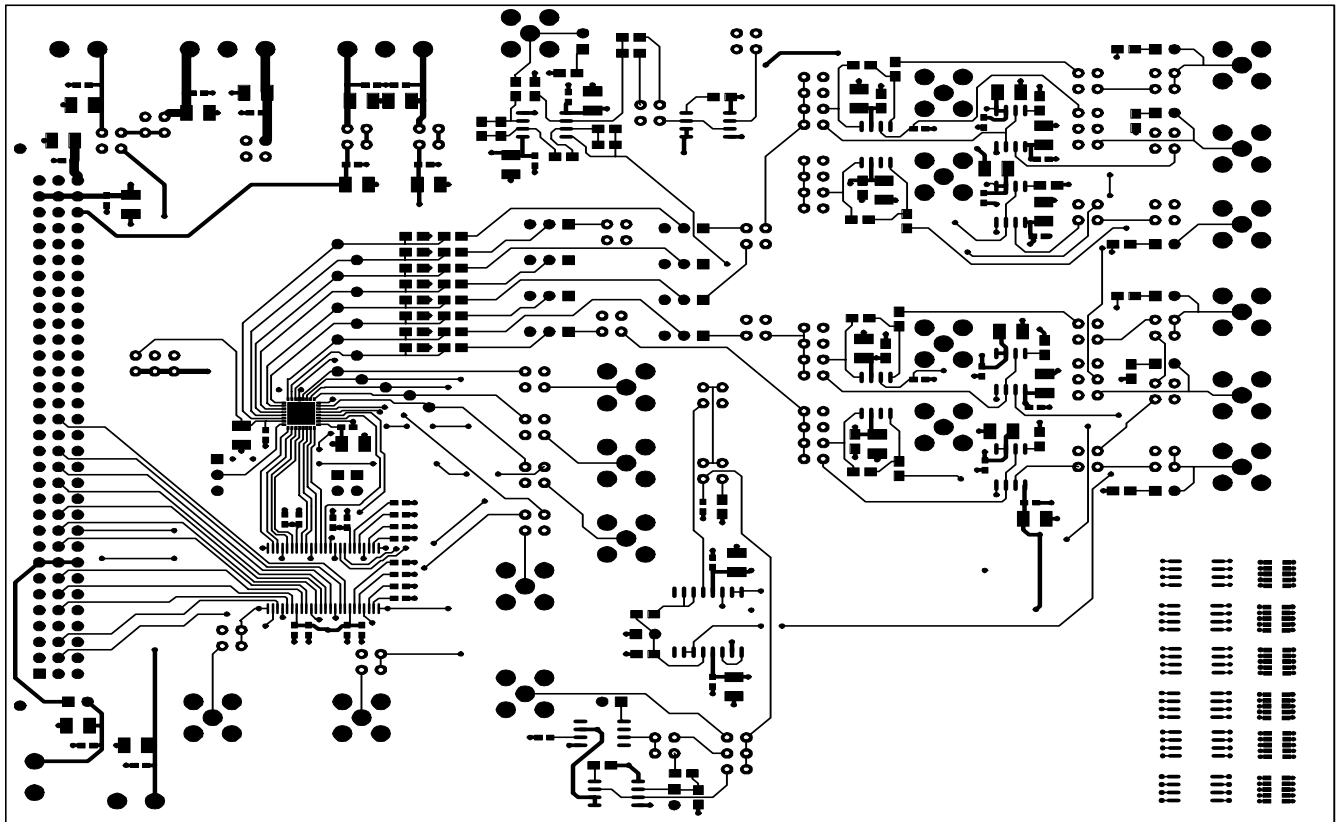


Figure 7: AD7938/39/38-6 Evaluation Board Layout, Component Side Artwork

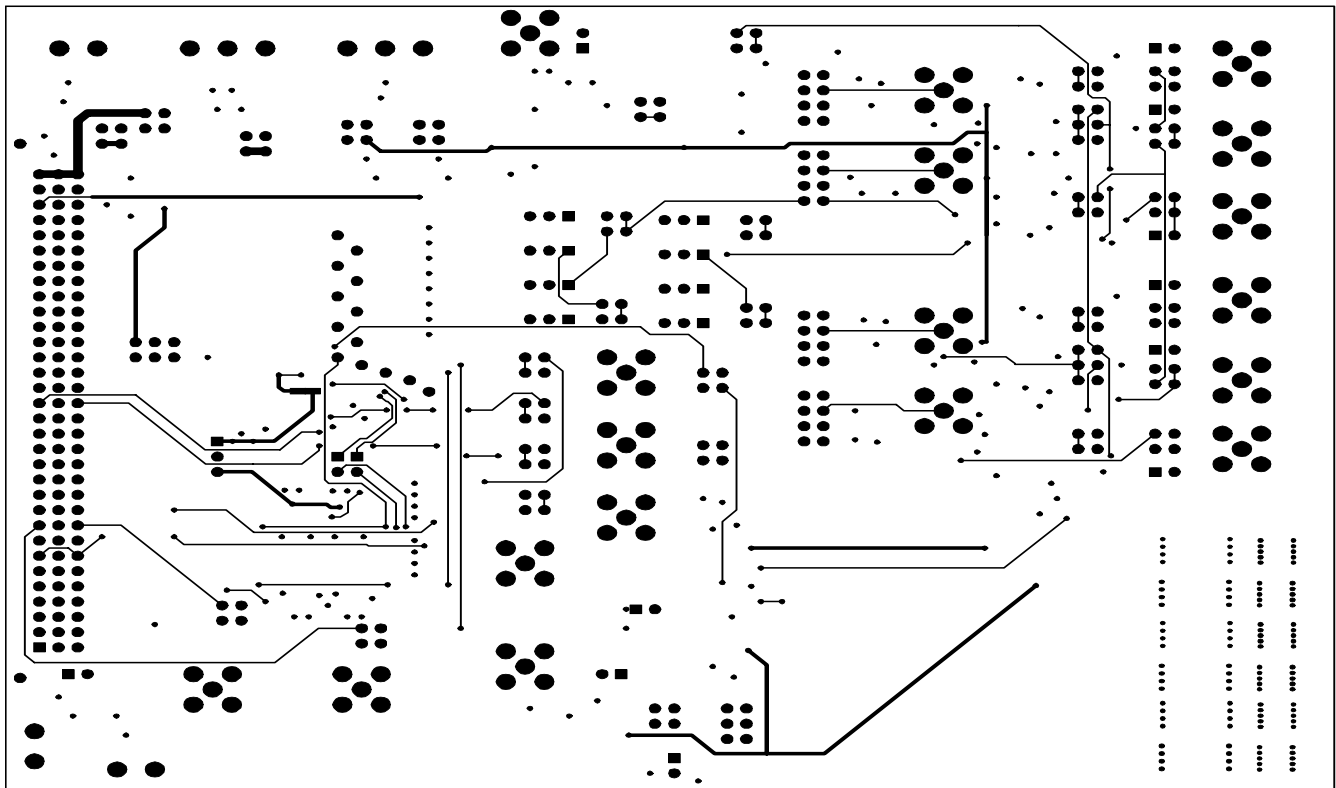


Figure 8: AD7938/39/38-6 Evaluation Board Layout, Solder Side Artwork

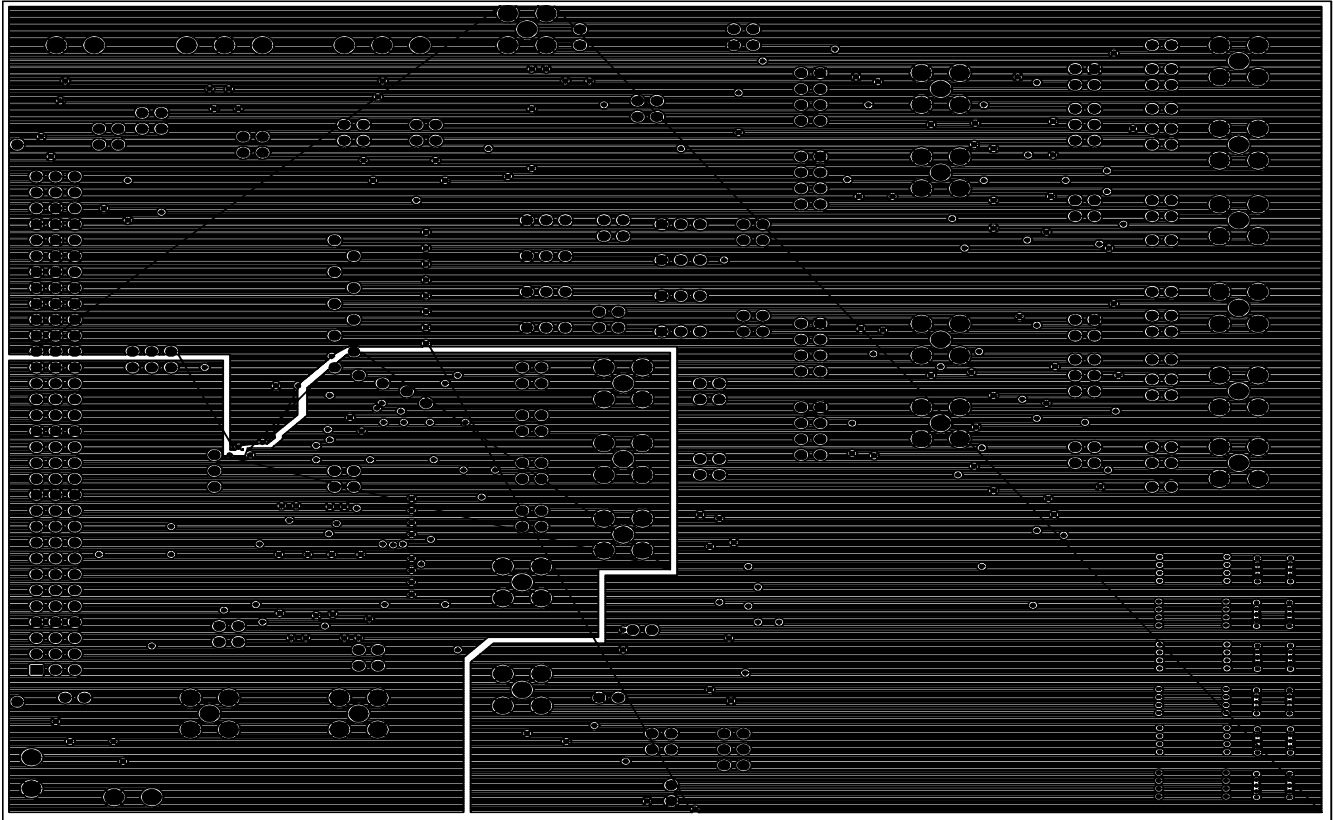


Figure 9: AD7938/39/38-6 Evaluation Board Layout, Layer 2

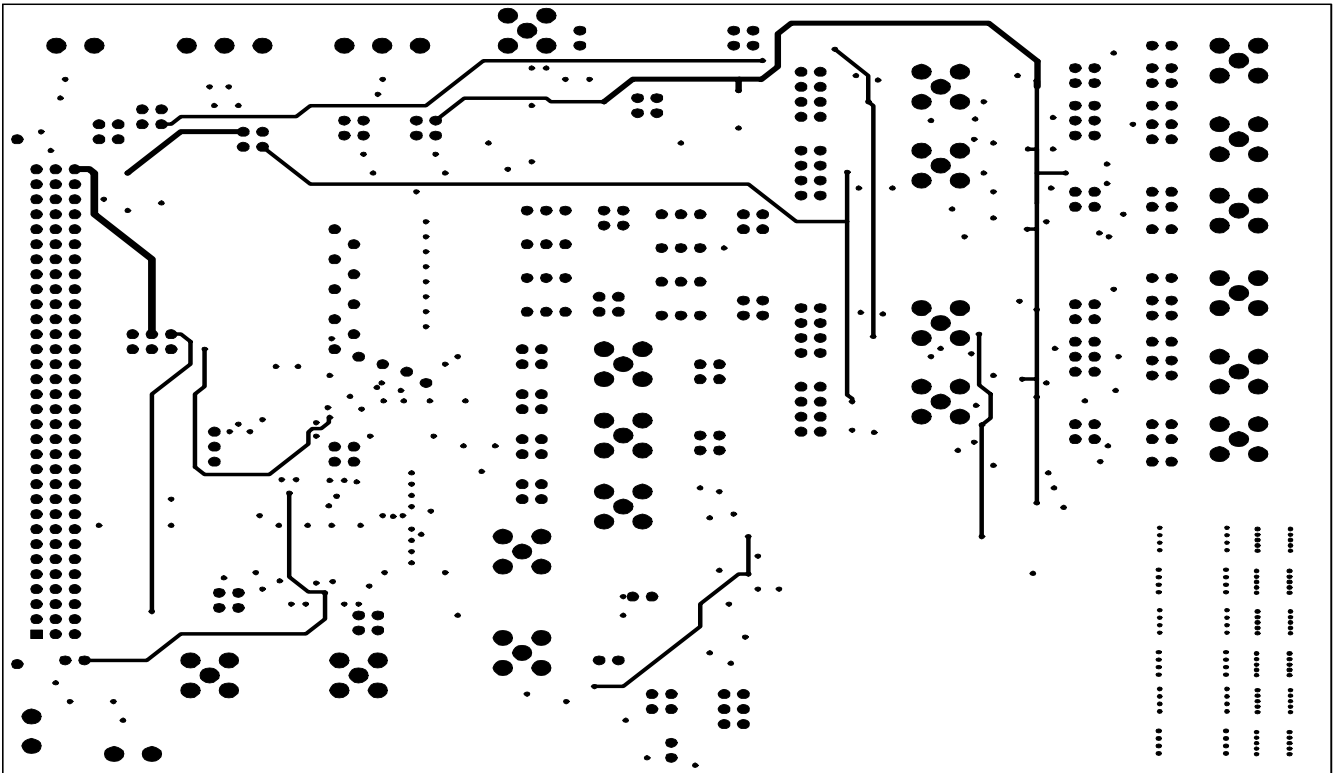


Figure 10: AD7938/39/38-6 Evaluation Board Layout, Layer 3

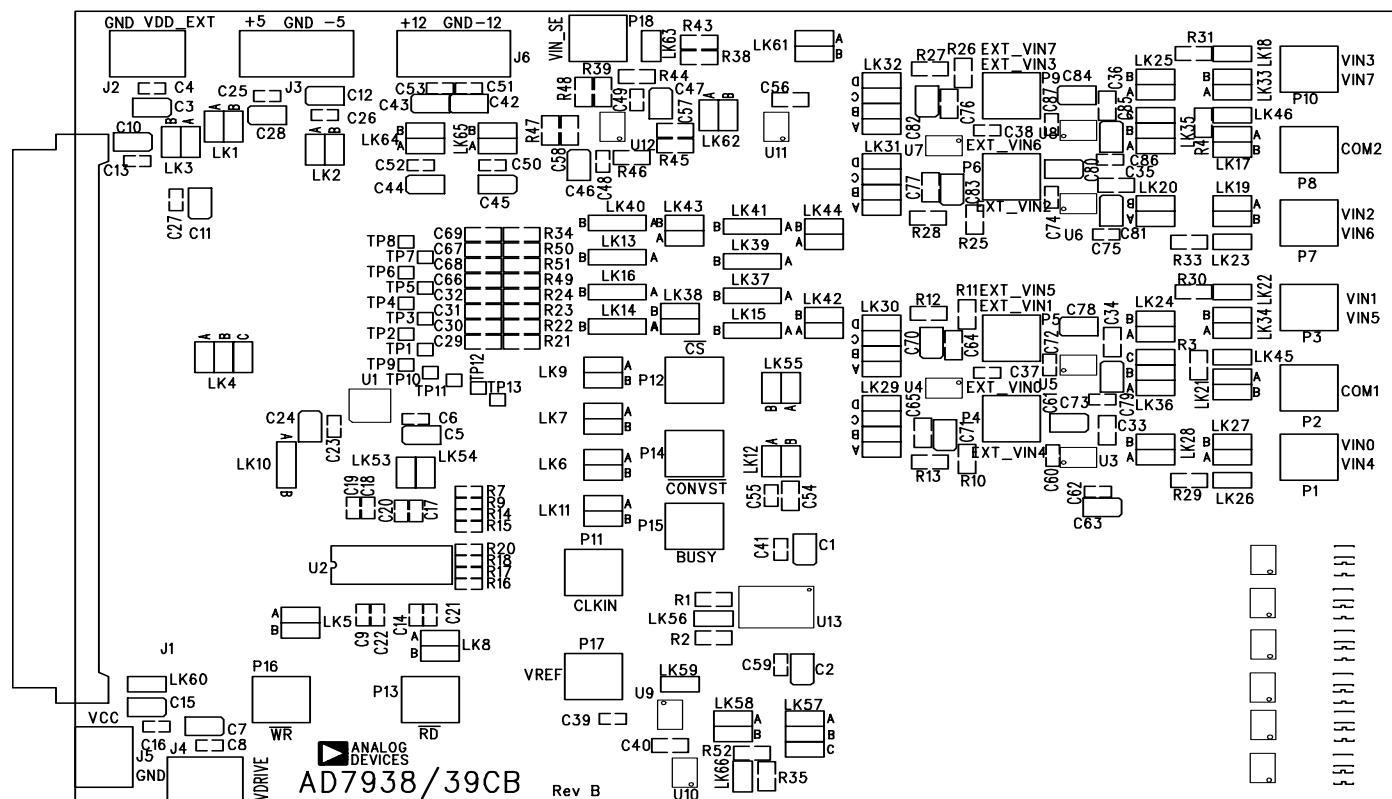


Figure 11: AD7938/39/38-6 Evaluation Board Layout, Silkscreen

Table VII – Bill Of Materials.

Quantity	Part Type	Reference Designator	Manuf / Disti	Order Number
1	AD7938/39	U1	ADI	AD7938BCP/AD7939BCP
1	74FCT162245	U2		IDT74FCT162245TPV
6	AD8021AR	U3, U5, U6, U8, U10, U11	ADI	AD8021AR
2	AD8138	U4, U7	ADI	AD8138AR
1	AD780	U9	ADI	AD780AN/AR
1	AD8022	U12	ADI	AD8022AR
1	AD713	U13	ADI	AD713JR-16
28	10uF Tantalum Capacitor	C1-C3, C5, C7, C10-C12, C15, C24, C28, C42-C47, C61, C63, C70, C71, C78-C85		FEC 197-427
39	0.1uF Ceramic Capacitor SMD 0603	C4, C6, C8, C9, C13, C14, C16-C23, C25-C27, C37-C39, C41, C48-C53, C55, C59, C60, C62, C64, C65, C72-C75, C86, C87		FEC 499-675
2	0.1uF Ceramic Capacitor SMD 0805	C76, C77		FEC 317-287
8	10nF Ceramic Capacitor SMD 0805	C29-C32, C66-C69		FEC 499-225
6	10pF Ceramic Capacitor SMD 0805	C33-C36, C40, C56		FEC 344-8952
1	470nF Ceramic Capacitor SMD 0805	C54		FEC 755-620
2	68pF Ceramic Capacitor SMD 0805	C57, C58		FEC 722-066
7	51Ohm, 0.1W, 0.1% Resistor SMD 0805	R3, R4, R29-R31, R33, R44	Multicomp	FEC 321-7905
7	62Ohm, 0.1W, 0.1% Resistor SMD 0805	R21-R24, R49-R51	Multicomp	FEC 321-7917
1	100Ohm, 0.1W, 0.1% Resistor SMD 0805	R52	Multicomp	FEC 911-732
2	300Ohm, 0.1W, 0.1% Resistor SMD 0805	R1, R2	Multicomp	FEC 771-272
1	390Ohm, 0.1W, 0.1% Resistor SMD 0805	R35	Multicomp	FEC 613-046
8	499Ohm, 0.1W, 0.1% Resistor SMD 0805	R10-R13, R25-R28	Multicomp	FEC 553-712
6	1kOhm, 0.1W, 0.1% Resistor SMD 0805	R39, R43, R45-R48	Multicomp	FEC 613-095
1	3kOhm, 0.1W, 0.1% Resistor SMD 0806	R38	Multicomp	FEC 771-399
8	100kOhm, 0.1W, 0.1% Resistor SMD 0805	R7, R9, R14-R18, R20	Multicomp	FEC 612-728
14	TESTPOINT	TP1-T13	W-Hughes	FEC 240-333
19	Gold 50Ohm SMB Jack	P1-P18	M/A-Com	FEC 310-682
1	CON41612\96 Connector	J1	Siemens	FEC 225-393
3	2 Way Terminal block	J2, J4, J5		FEC 304-4889
2	3 Way Terminal Block	J3, J6		FEC 304-4890
	JUMPERS	LK1-LK66	Harwin	FEC 511-705
66	Shorting Link	LK1-LK66	Berg	FEC 150-411
4	Stick On Feet	Each Corner	3M	FEC 148-922